



K. S. INSTITUTE OF TECHNOLOGY, BENGALURU-560109
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

PROJECT PHASE - I (BCS685)

PROJECT TITLE: OncoBot: Explainable AI for Bladder Cancer Survival Prediction

Guide's Name with Signature: Mr. Gowtham Raj M

Batch No.: 2026_CSE_16

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ABSTRACT

Cancer prognosis and treatment planning require the analysis of multiple types of biomedical data, particularly **genomic information and histopathological tissue images**. Gene expression data provides insights into the **molecular mechanisms involved in cancer**, while histopathological images reveal the **structural and morphological characteristics of tumor tissues**. Traditionally, these data sources are analyzed independently, which limits the ability to understand the relationship between **genetic activity and tumor morphology**.

Recent developments in **deep learning and artificial intelligence** have enabled automated analysis of complex medical datasets. However, many existing deep learning models used in cancer prediction operate as **black-box systems**, providing limited insight into how predictions are generated. This lack of transparency makes it difficult for clinicians and researchers to trust and adopt such systems in real-world healthcare environments where **interpretability and reliability are essential**.

This project proposes **OncoBot**, a multimodal explainable artificial intelligence framework designed for **bladder cancer prognosis**. The system integrates **gene expression data and histopathological whole-slide images** to learn relationships between molecular activity and tumor morphology. The core of the system is based on a **Pathway-Aware Multimodal Transformer (PAMT)** architecture, which models interactions between **biological pathways derived from gene expression data and visual features extracted from histopathology images**. Gene expression values are mapped to biologically meaningful pathways using the **KEGG pathway database**, while histopathology images are processed through image feature extraction techniques to obtain representative tissue embeddings.

To improve interpretability, the system incorporates a **neuro-symbolic explanation module** that combines attention-based outputs from the neural model with structured biological knowledge. This module generates **visual heatmaps on histopathology images**, highlighting tumor regions that



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contribute to the prediction, and produces **structured explanations linking important biological pathways with observed tumor features**.

The proposed system will be evaluated using publicly available datasets from **The Cancer Genome Atlas (TCGA)** and histopathology image repositories such as **The Cancer Imaging Archive (TCIA)**. Evaluation will focus on both **prediction performance and interpretability**, assessing the model's ability to estimate survival risk while providing meaningful explanations.

The goal of this project is to develop a **transparent and computationally efficient AI framework** that integrates genomic and imaging data for cancer prognosis. By combining multimodal learning with explainable AI techniques, the system aims to support research in **AI-driven biomedical analysis and interpretable machine learning for healthcare applications**.

System Requirements(H/W and S/W)

Minimum Hardware Requirements:

1. **Processor:** Intel Core i5 / AMD Ryzen 5 or higher
2. **RAM:** 8 GB (16 GB recommended for model training)
3. **Storage:** 256 GB SSD
4. **GPU (Recommended):** NVIDIA CUDA-enabled GPU (for faster training)
5. **Display:** Standard monitor (for visualization of explanations)

Software Requirements:

1. **Operating System:** Windows / Linux
2. **Programming Language:** Python 3.x
3. **Deep Learning Frameworks:** TensorFlow / PyTorch
4. **Machine Learning Libraries:** Scikit-learn, NumPy, Pandas
5. **Explainability Libraries:** Captum / SHAP / Grad-CAM implementations
6. **Symbolic Reasoning Tools:** Rule-based inference (custom logic / knowledge graphs)
7. **Development Tools:** Jupyter Notebook, VS Code
8. **Datasets:** Public medical datasets (TCGA, TCIA, CAMELYON, ISIC, etc.)

Base Paper Submitted: (Yes/No)

1. Rui Yan, Xueyuan Zhang, Zihang Jiang, Baizhi Wang, Xiuwu Bian, Fei Ren, S. Kevin Zhou, "**Pathway-Aware Multimodal Transformer (PAMT): Integrating Pathological Image and Gene Expression for Interpretable Cancer Survival Analysis**," *IEEE Transactions on Pattern Analysis and Machine Intelligence* (January 2026), **Page no. 896–909**, Published by **IEEE**.



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Conference Papers :

- [1] Ribeiro, M. T., Singh, S., Guestrin, C., **“Why Should I Trust You? Explaining the Predictions of Any Classifier,”** Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (Aug 2016), Page no. 1135–1144, Published by ACM
- [2] Lundberg, S. M., Lee, S. I., **“A Unified Approach to Interpreting Model Predictions,”** Advances in Neural Information Processing Systems (NeurIPS) (Dec 2017), Page no. 4765–4774, Published by Curran Associates Inc.
- [3] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D, **“Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization,”** Proceedings of the IEEE International Conference on Computer Vision (Oct 2017), Page no. 618–626, Published by IEEE.
- [4] Chaddad, A., Peng, J., Xu, J., Bouridane, A., **“Survey of Explainable Artificial Intelligence Techniques in Healthcare,”** Sensors (Feb 2023), Page no. 1–38, Published by MDPI.

Books:

1. **“The Biology of Cancer”** By Robert A. Weinberg

Note: Not For Student Use

ACCEPTED	REJECTED	RE SUBMIT
Reason for Rejection:		
Reason for Re Submit:		

Project Coordinators

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